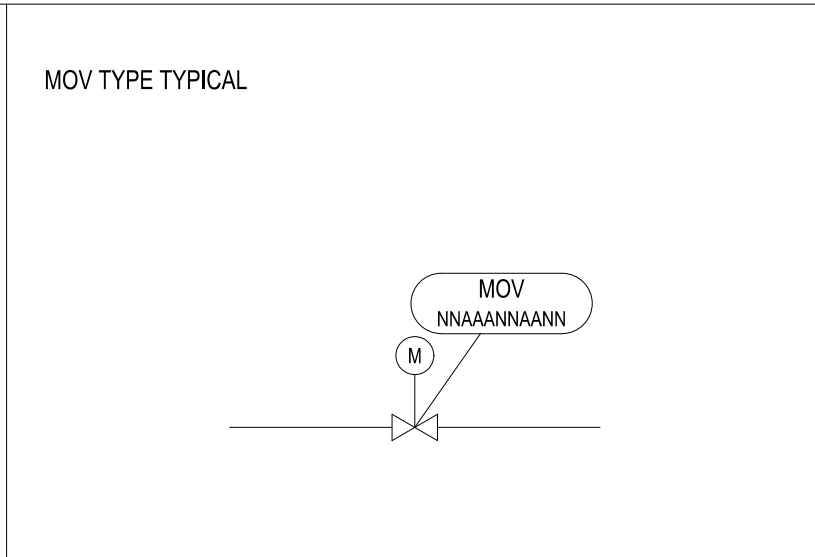
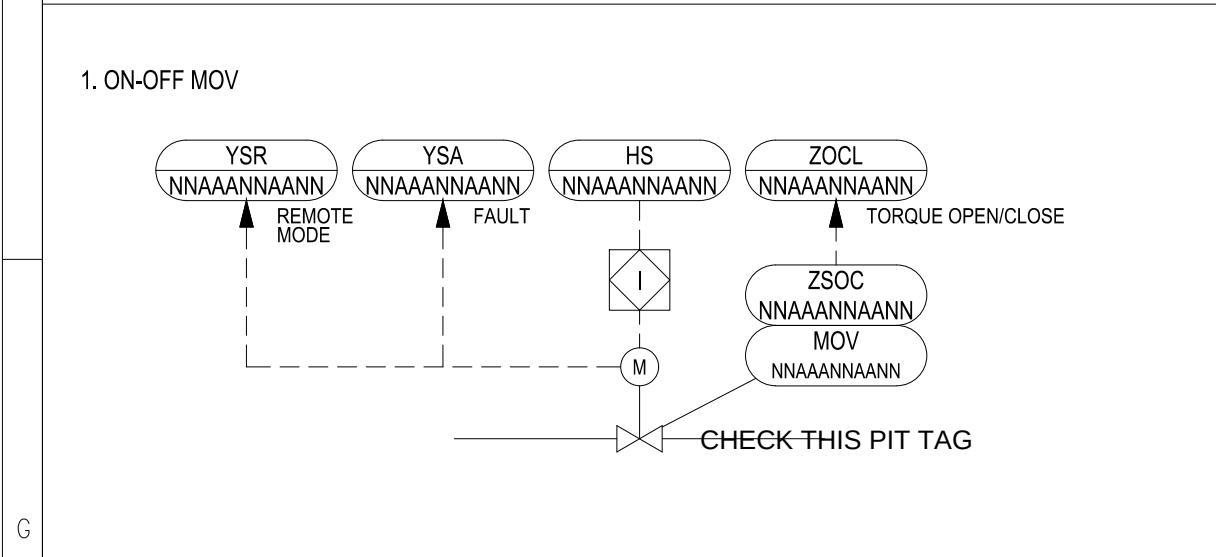


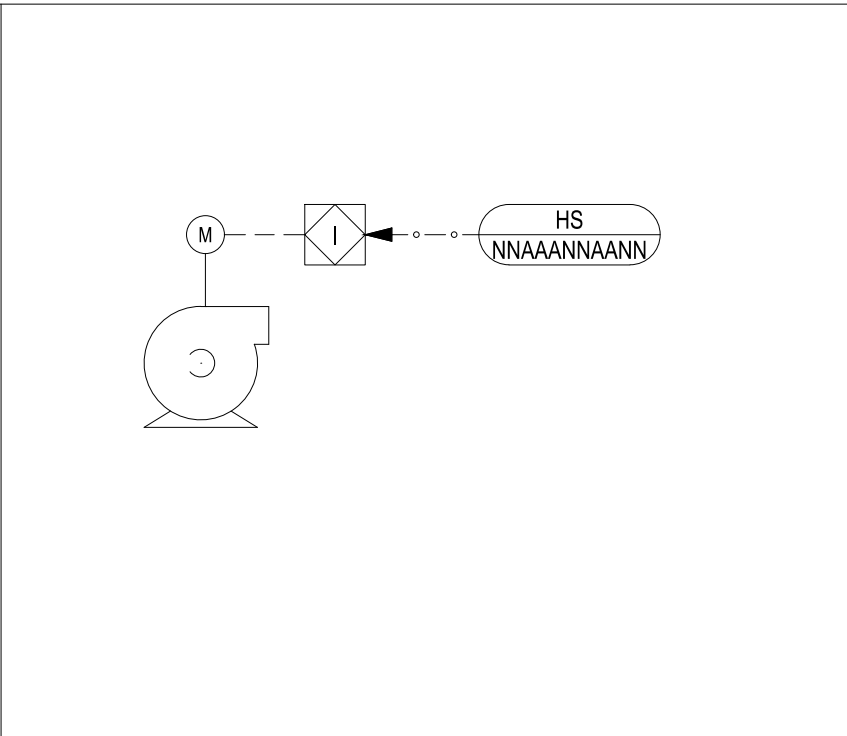
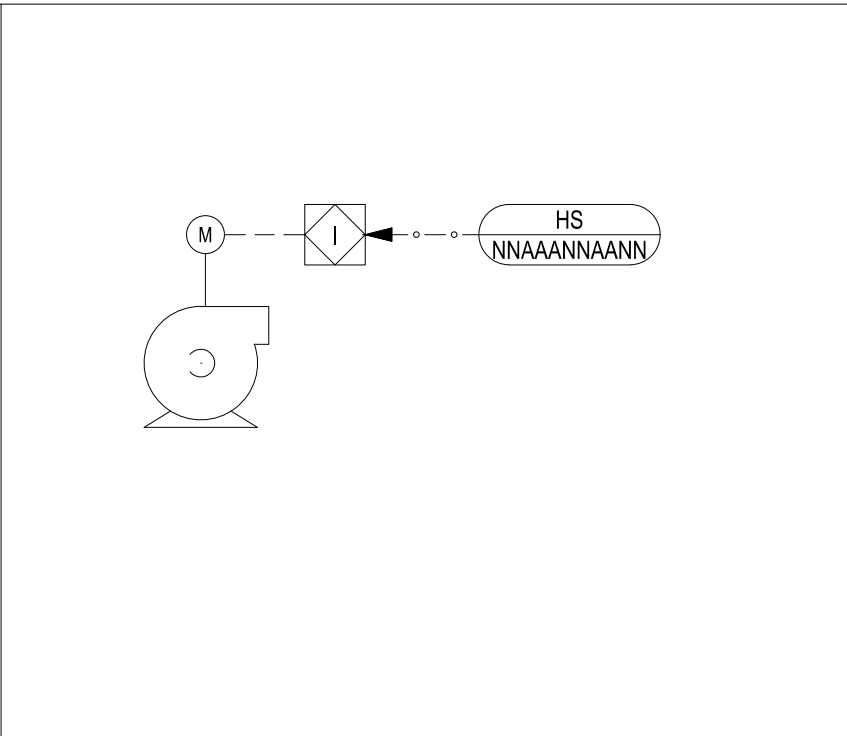


8		7		6		5		4		3		2		1													
FIRST LETTER		SUCCEEDING LETTERS														SELF-ACTUATED DEVICES-PRESSURE		GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.									
		SENSING DEVICE		DISPLAY DEVICE			CONTROL DEVICE			MISCELLANEOUS DEVICE																	
SYMBOL	MEASURED OR INITIATING VARIABLE	PRIMARY ELEMENT	TRANSMITTER	INDICATOR	RECORDER	INTEGRATING INDICATOR	ARARM ANNUNCIATOR			CONTROL STATION MAN/AUTO	CONTROLLER	CONTROL VALVE	OTHER FINAL CONTROL ELEMENT	SWITCH	LOCAL OBSERVATION GLASS	SOLENOID RELAY OR POSITIONER	TEST POINT	USER'S CHOICE	UNCLASSIFIED								
()	TYPICAL SYMBOL	()E	()T	()I	()R	()QI	LOW	HIGH	HIGH & LOW	()K	()C	()CV	()Z	()S	()G	()Y	()P	()N	()X								
A	ANALYSIS	A E	A T	A I	A R		AAL	AAH	AAHL	A K	A C		A Z	A S		A Y	A P	A N	A X								
B	BURNER FLAME	B E	B T	B I	B R		BAL				B C			B S		B Y	B P	B N	B X								
C	CONDUCTIVITY	C E	C T	C I	C R		CAL	CAH	CAHL	C K			C Z	C S		C Y	C P	C N	C X								
D	DENSITY (MASS) OR SPECIFIC GRAVITY	D E	D T	D I	D R		DAL	DAH	DAHL	D K	D C		D Z	D S		D Y	D P	D N	D X								
E	VOLTAGE (EMF)	E E	E T	E I	E R		EAL	EAH	EAHL	E K	E C		E Z	E S		E Y		E N	E X								
F	FLOW	F E	F T	F I	F R	FQI	FAL	FAH	FAHL	F K	F C	FCV	F Z	F S	F G	F Y	F P	F N	F X								
G	GAUGING (DIMENSIONAL)	G E	G T	G I	G R		GAL	GAH	GAHL	G K	G C			G S		G Y		G N									
H	HAND (MANUAL INITIATED)	H									H C		H Z	H S		H Y		H N									
I	CURRENT	I E	I T	I I	I R	IQI	IAL	IAH	IAHL	I K			I Z	I S		I Y		I N	I X								
J	POWER	J E	J T	J I	J R	JQI	JAL	JAH	JAHL	J K			J Z	J S		J Y		J N	J X								
K	TIME	K E	K T	K I	K R	KQI	KAL	KAH	KAHL		K C		K Z	K S		K Y		K N	K X								
L	LEVEL	L E	L T	L I	L R		LAL	LAH	LAHL	L K	L C	LCV	L Z	L S	L G	L Y	L P	L N	L X								
M	MOISTURE OR HUMIDITY	M E	M T	M I	M R		MAL	MAH	MAHL	M K	M C		M Z	M S		M Y	M P	M N	M X								
O	USER'S CHOICE																										
P	PRESSURE OR VACUUM	P E	P T	P I	P R		PAL	PAH	PAHL	P K	P C	PCV	P Z	P S		P Y	P P	P N	P X								
PD	PRESSURE DIFFERENTIAL	PDE	PDT	PDI	PDR		PDAL	PDAH	PDAHL	PDK	PDC		PDZ	PDS													
Q	QUANTITY	Q E	Q T	Q I	Q R	QQI	QAL	QAH	Q AHL	Q K	Q C		Q Z	Q S		Q Y		Q N	Q X								
S	SPEED OR FREQUENCY	S E	S T	S I	S R	SQI	SAL	SAH	SAHL	S K	S C		S Z	S S		S Y		S N	S X								
T	TEMPERATURE	T E	T T	T I	T R		TAL	TAH	TAHL	T K	T C	TCV	T Z	T S		T Y	T P	T N	T X								
U	MULTIVARIABLE			U I	U R								U Z	U S		U Y		U N	U X								
V	VIBRATION		V T	V I	V R		VAL	VAH	VAHL	V K	V C		V Z	V S		V Y	V P	V N	V X								
W	WEIGHT OR FORCE	W E	W T	W I	W R	WQI	WAL	WAH	WAHL	W K	W C		W Z	W S		W Y		W N	W X								
X	UNCLASSIFIED																										
Y	EVENT OR STATE	Y E	Y T	Y I	Y R		YAL	YAH					Y Z	Y S		Y Y		Y N	Y X								
Z	POSITION	Z E	Z T	Z I	Z R		ZAL	ZAH	Z AHL				ZZ	Z S		Z Y		Z N	Z X								
GENERAL INSTRUMENTS				SENSOR-FLOW				SENSOR-FLOW (CONT.)				SENSORS-PRESSURE				VALVES ACTUATORS											
SIGNALS & LINES								SENSOR-LEVEL				SENSORS-TEMPERATURE															
								SENSOR-ANALYSIS				VALVE BODIES & FAILURE MODES															
GENERAL NOTES :		1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.																									
NOTES :																											
SELF-ACTUATED DEVICES-FLOW																											
SELF-ACTUATED DEVICES-LEVEL																											
FOR INTERNAL USE																											
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TENDERER		SAMSUNG SAMSUNG C&T Engineering & Construction Group																									
PROJECT NAME		AL NOUF1 PROJECT																									
DRAWING TITLE		P&ID FOR SYMBOLS & LEGENDS 2																									
PROJECT DWG NO.		D00P-00GEN00-M05-0003																									
PROJECT NO.		D00P																									
DATE		19.SEP.2025																									
DESCRIPTION		FOR INTERNAL USE																									
PREPARED																											
REVIEWED																											
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SHEET		1/1																									
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REF. CODE																											

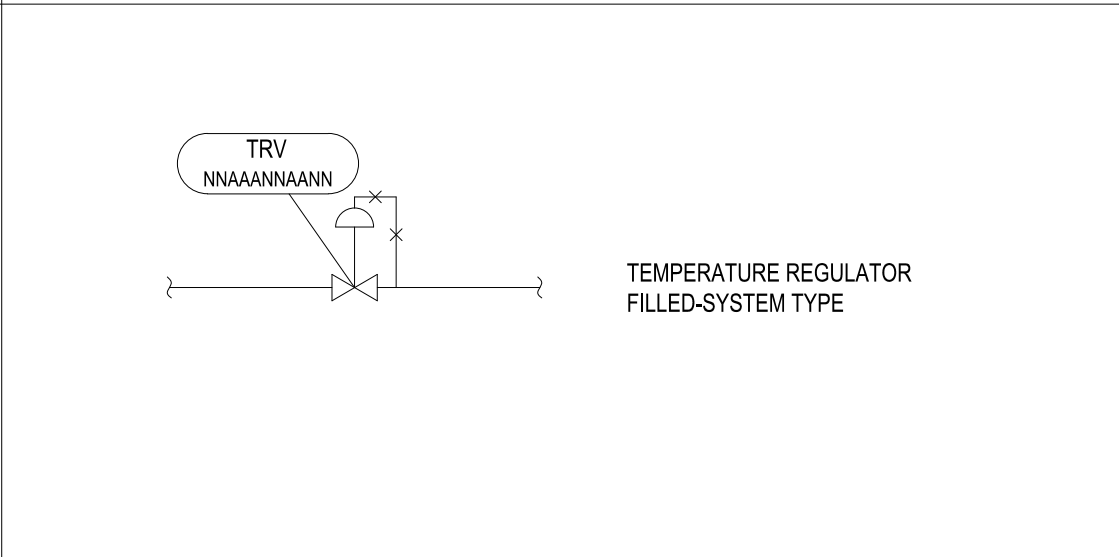
	8	7	6
H	VALVE BODY WITH ACTUATOR-TYPICAL FOR MOTOR OPERATED VALVES		



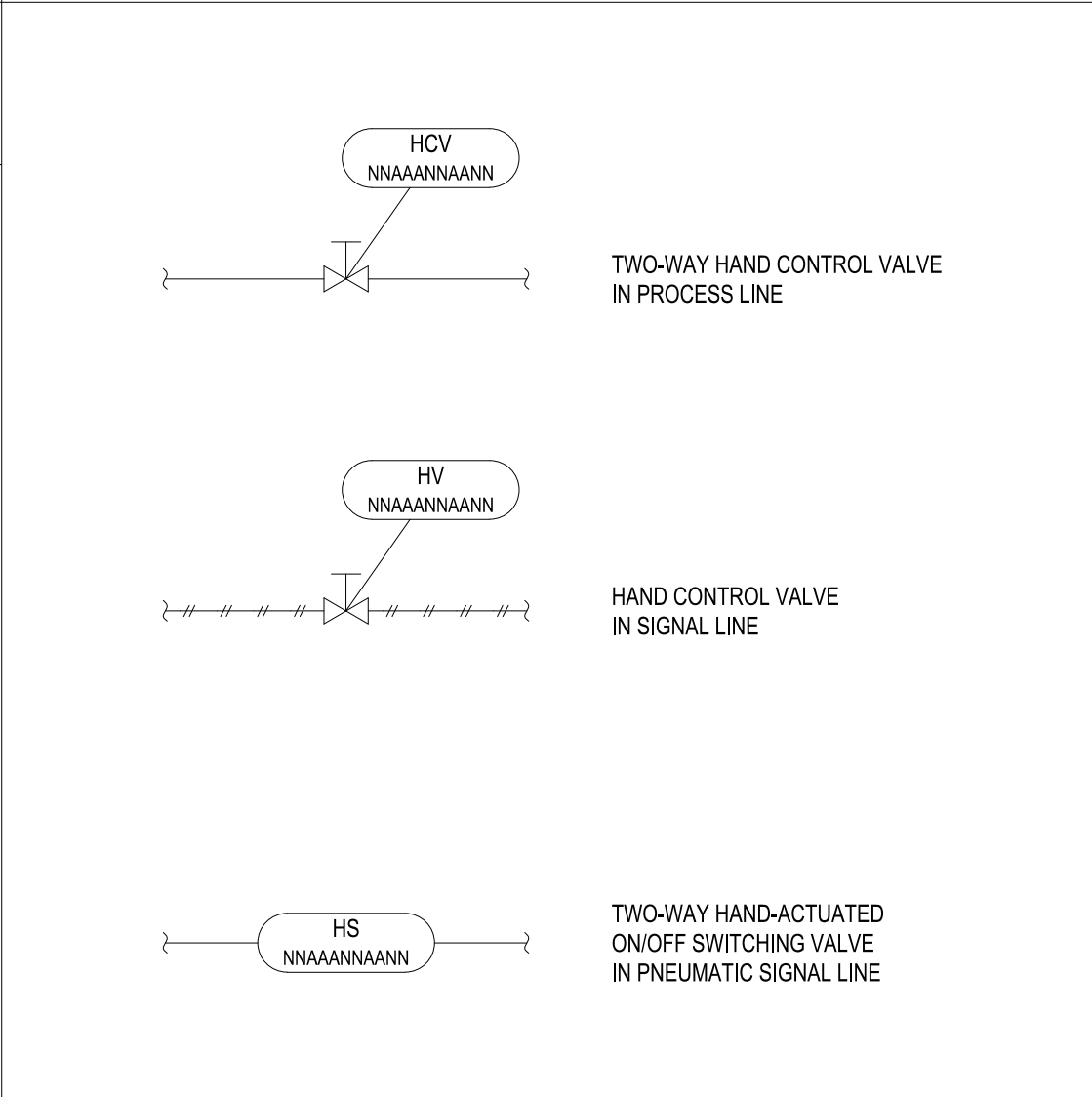
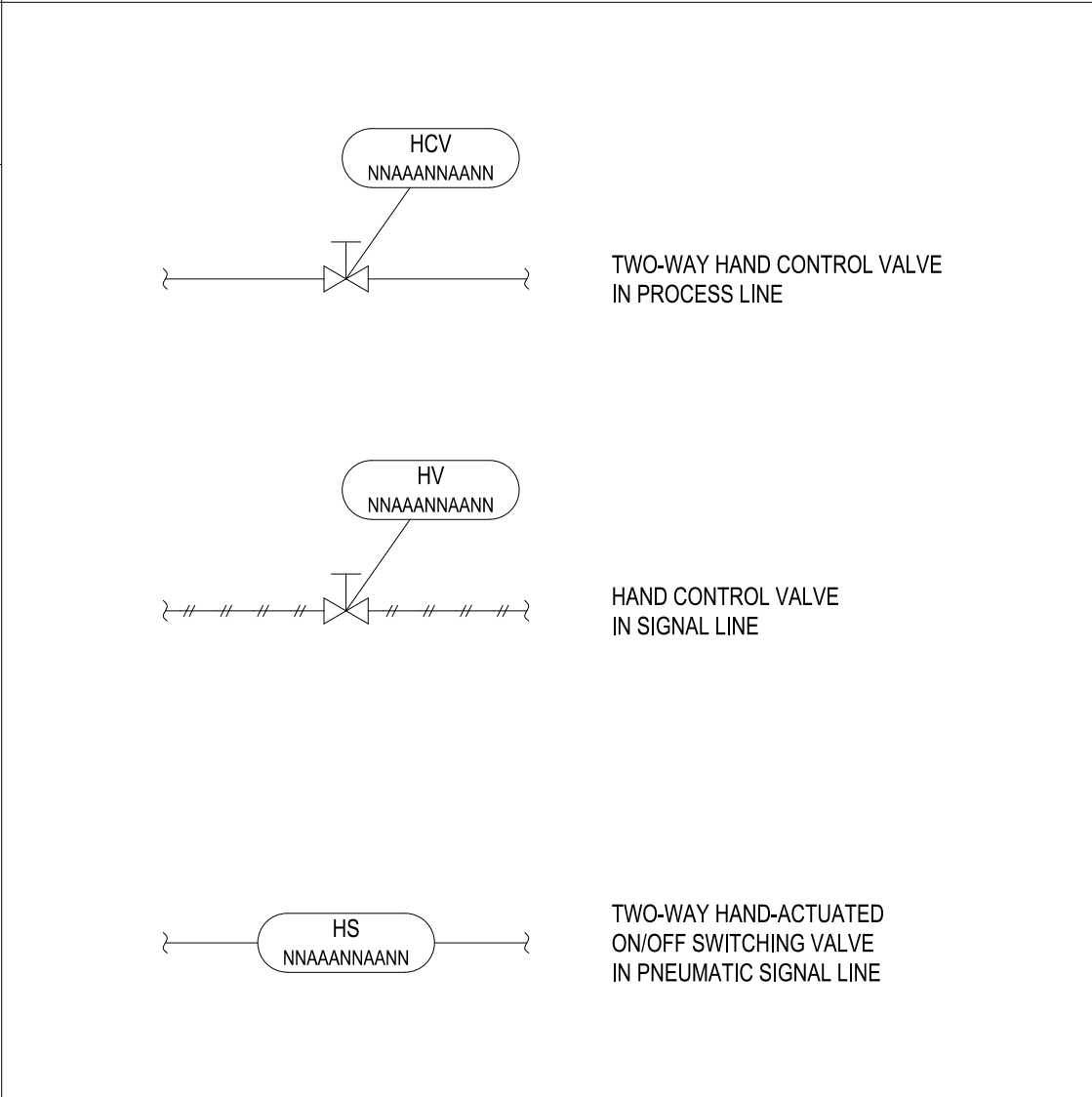
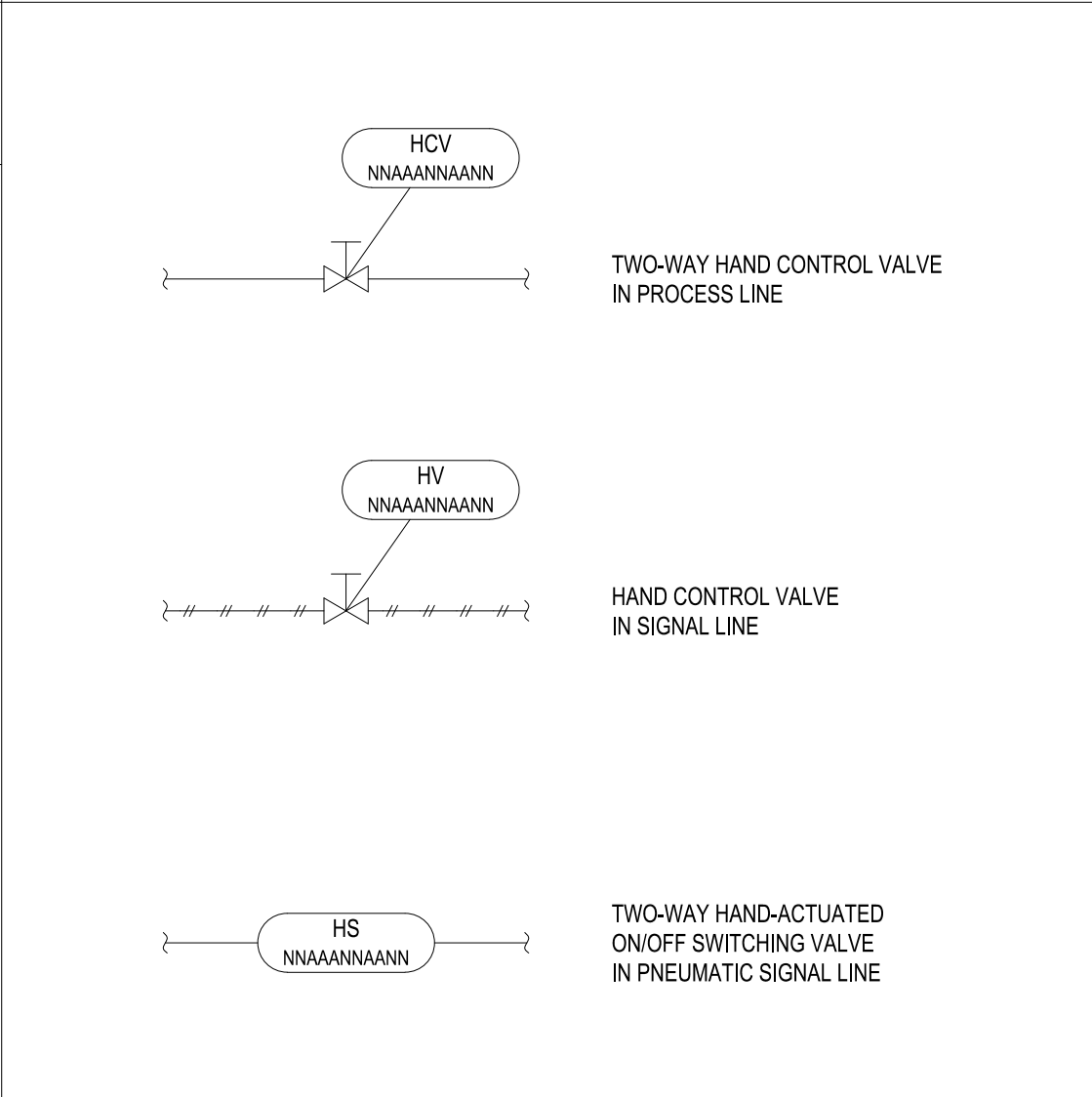
MOTOR WITHOUT DUTY/STANDBY SELECTION (TYP.1)



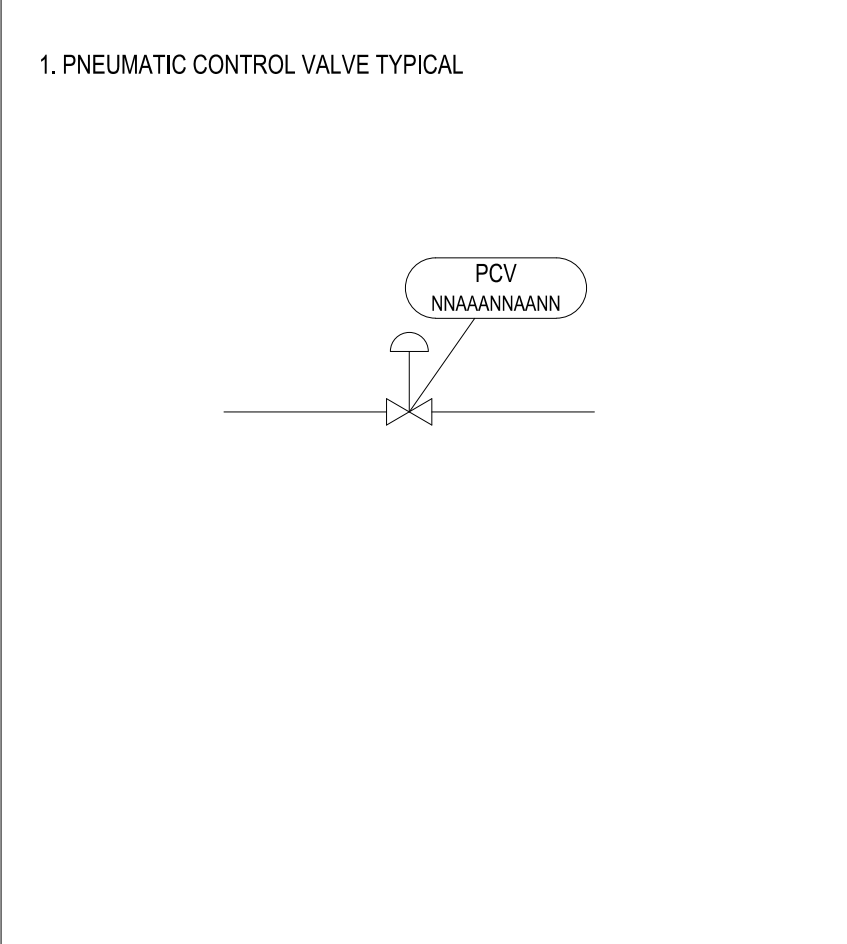
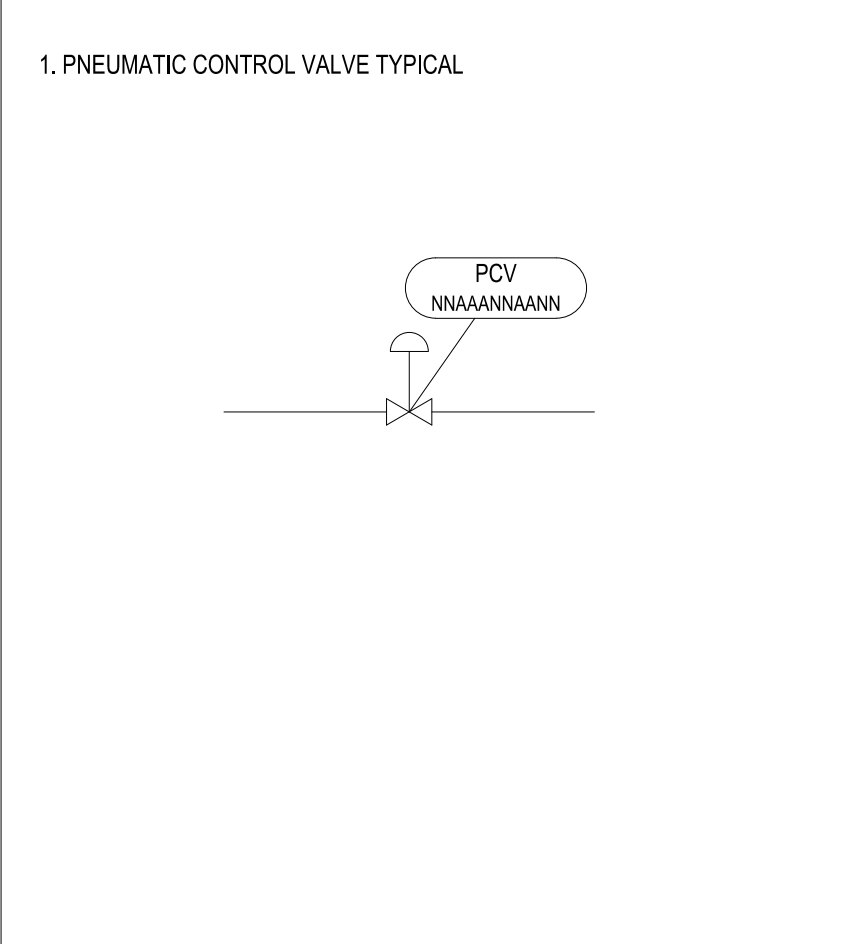
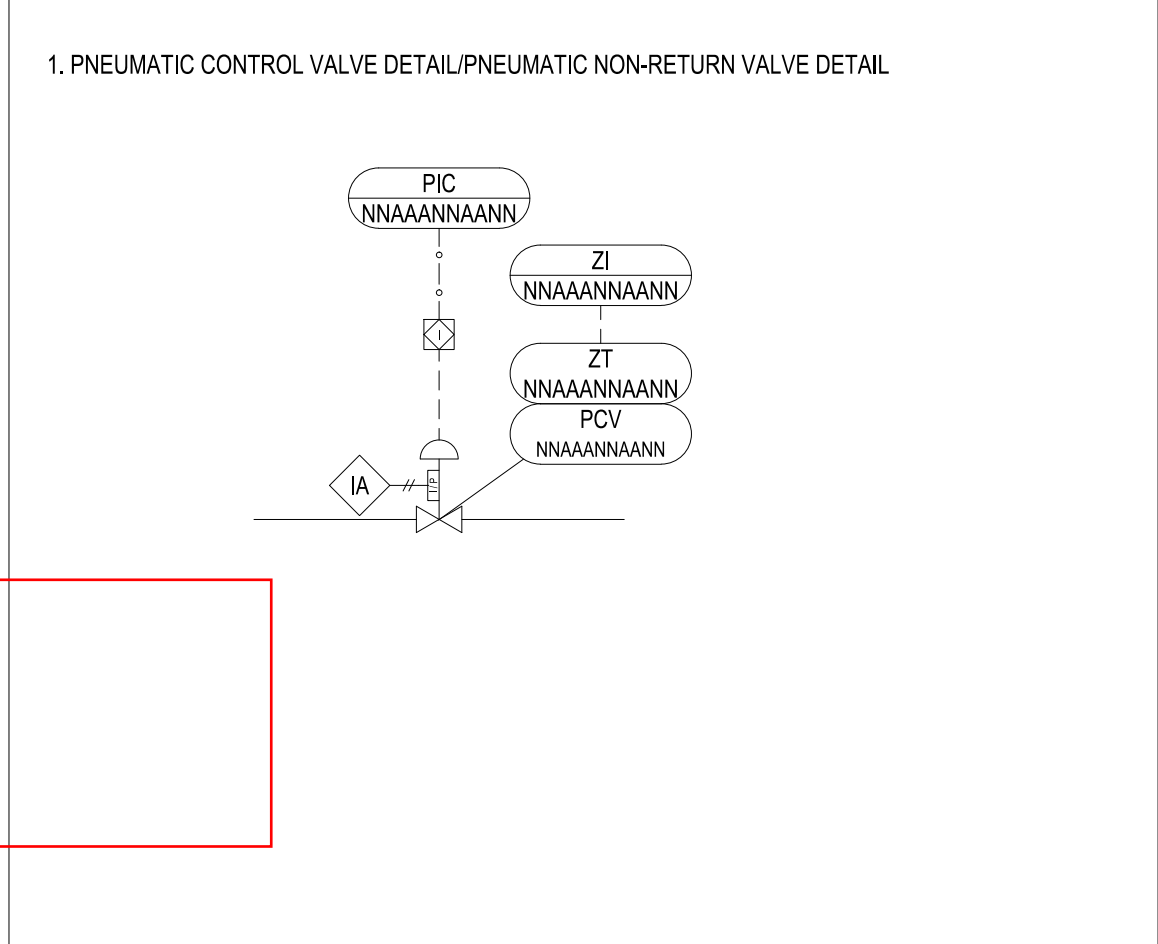
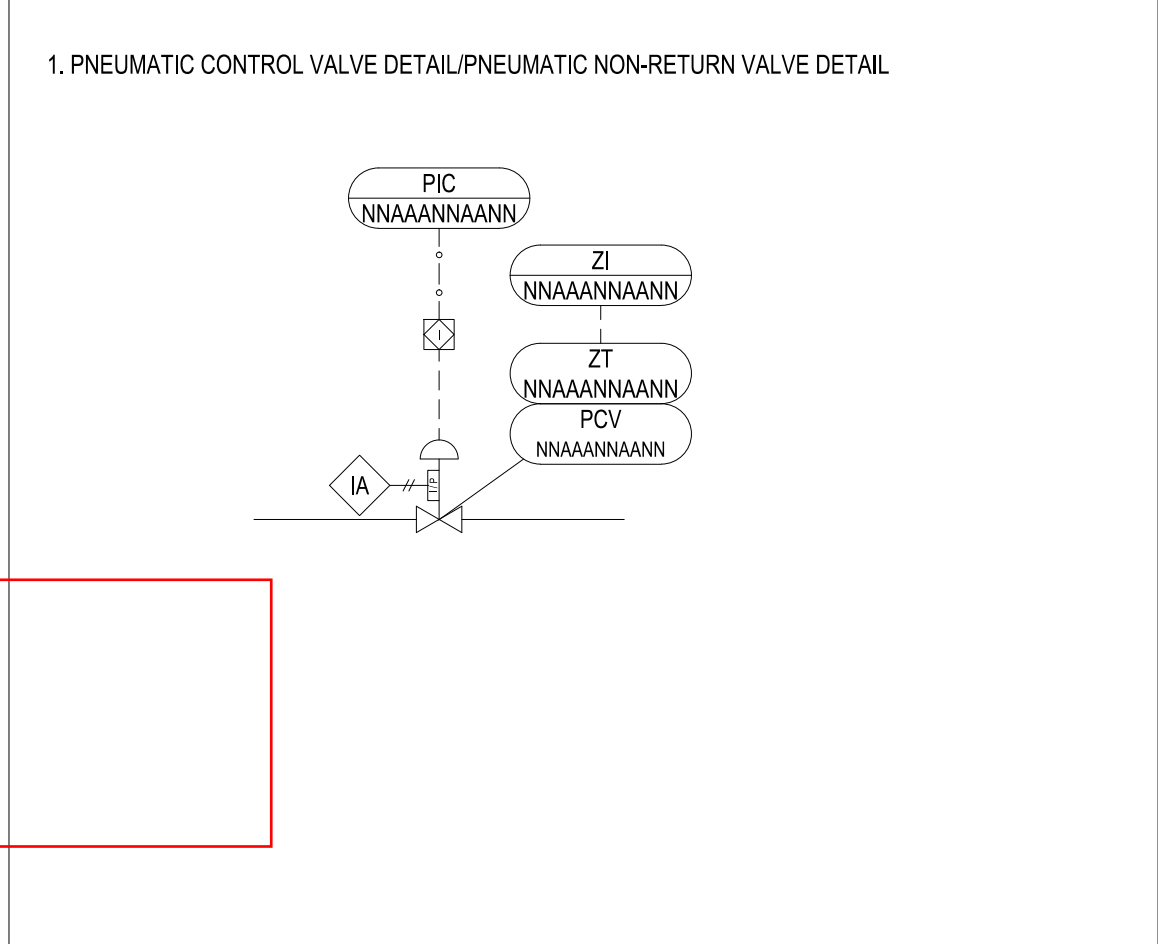
3	2
SELF-ACTUATED DEVICES-TEMP	



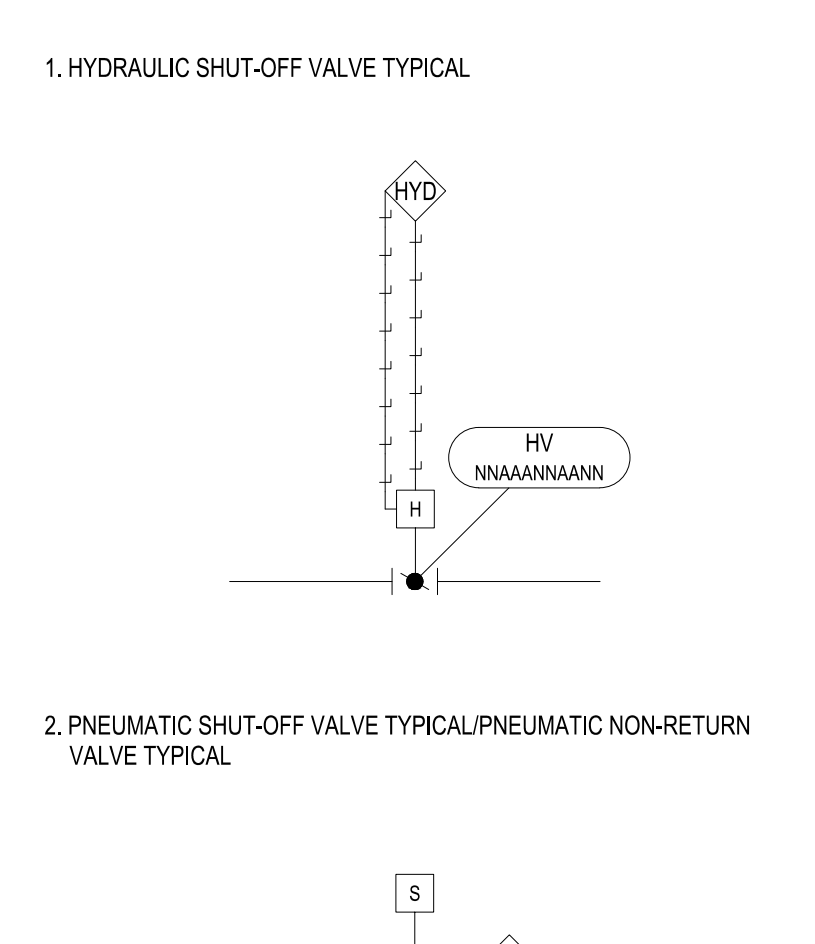
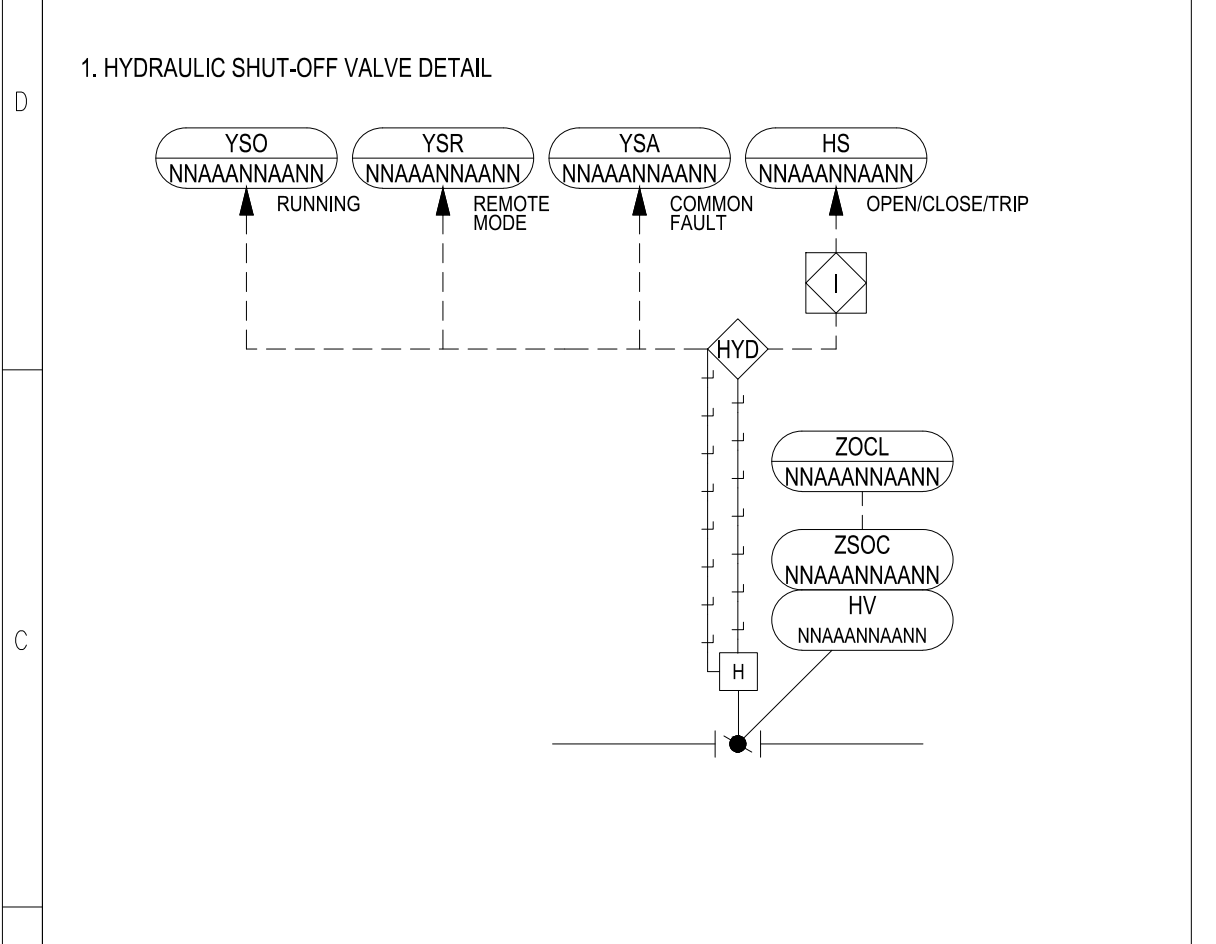
HAND ACTUATED DEVICES



CONTROL VALVE TYPICAL FOR PNEUMATIC OPERATED VALVES



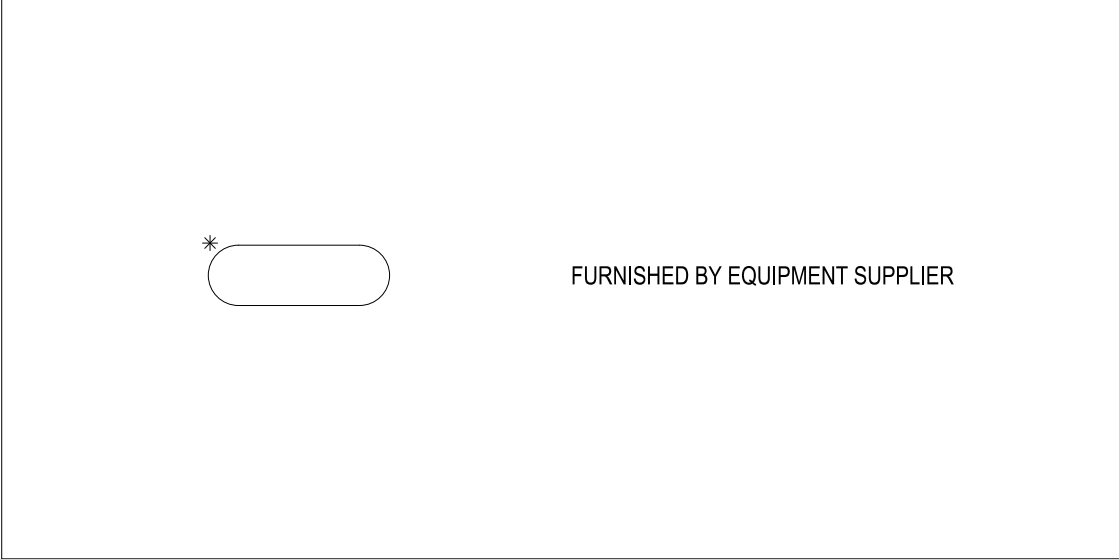
ON-OFF VALVE TYPICAL FOR HYDRAULIC AND PNEUMATIC OPERATED VALVES



ABBREVIATION

ESDV : EMERGENCY SHUT-DOWN VALVE
SDV : SHUT-DOWN VALVE
SET : SET VALUE
ZSOC : POSITION SWITCH OPEN/CLOSE
ZIOC : POSITION INDICATOR OPEN/CLOSE
ZOCL : POSITION OPEN/CLOSE LAMP
CC : CATIONIC CONDUCTIVITY
SC : SPECIFIC CONDUCTIVITY
XV : SHUT-OFF VALVE
M : ELECTRIC MOTOR
MH : MAN HOLE
SMFCP : SETTLEMENT METERING FLOW COMPUTER PANEL
BMFCP : BACK-UP METERING FLOW COMPUTER PANEL
MECP: MAIN FIRE ALARM CONTROL PANEL
ESEW : EMERGENCY SHOWER & EYE WASHER
DCMS : DISTRIBUTED CONTROL AND MONITORING SYSTEM
BMS : BURNER MANAGEMENT SYSTEM
BPS : BOILER PROTECTION SYSTEM

SUPPLIER DESIGNATOR



LOGIC & SEQUENCE CONTROL

	GENERAL - INTERCONNECTING LOGIC OR SEQUENCE CONTROL NORMALLY USED FOR LOCAL AUXILIARY CONTROL SYSTEM
	NOT NORMALLY ACCESSIBLE TO OPERATOR - PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROLS INTEGRAL TO THE DISTRIBUTED CONTROL EQUIPMENT
	NORMALLY ACCESSIBLE TO OPERATOR - PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROLS INTEGRAL TO THE DISTRIBUTED CONTROL EQUIPMENT

A	19.SEP.2025	FOR INTERNAL USE			
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					

TENDERER		 SAMSUNG C&T Engineering & Construction Group	
PROJECT NAME			
AL NOUF1 PROJECT			
DRAWING TITLE			
P&ID FOR SYMBOLS & LEGENDS 3			
PROJECT DWG. NO.		SHEET	REV.
		1/1	A
D00P-00GEN00-M05-0004			
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE
D00P		NONE	

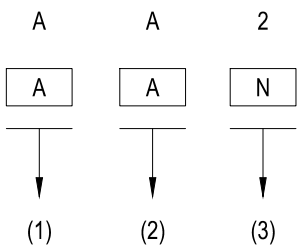
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DRAWING TITLE			
P&ID FOR SYMBOLS & LEGENDS 3			
PROJECT DWG NO.		SHEET	REV.
D00P-00GEN00-M05-0004		1/1	A
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE
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PROJECT DWG NO. D00P-00GEN00-M05-0004		SHEET	REV.
		1/1	A
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE
D00P		NONE	

TYPICAL IDENTIFICATION SYSTEM FOR
PIPE SIZE,
CLASS, MATERIAL AND INSULATION

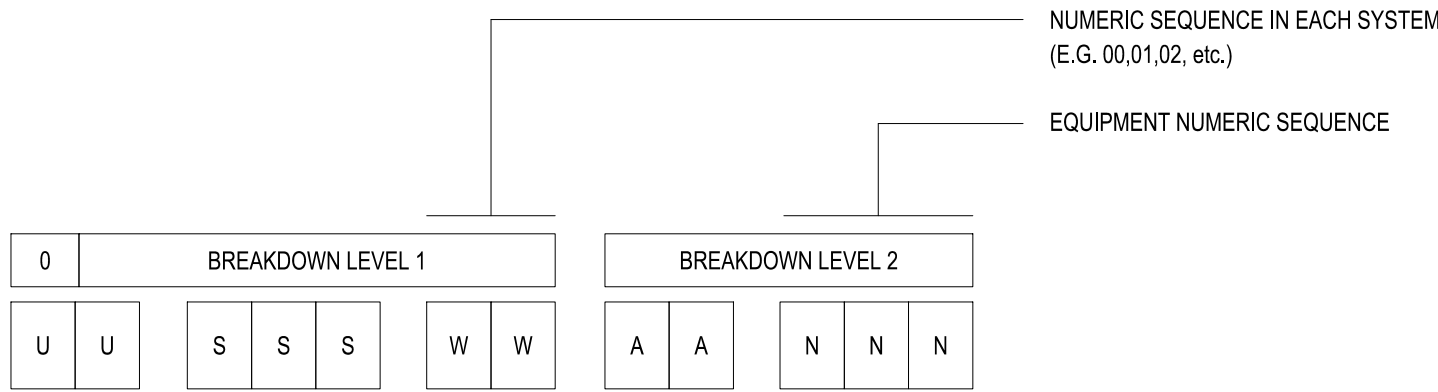
EXAMPLE OF
APPLICATION :



- (1) CLASSRATING
A : 4500# or 2500#SP (ASME B16.5)
B : 2500# (ASME B16.5)
C : 1500#SP (ASME B16.5)
D : 1500# (ASME B16.5)
E : 900# (ASME B16.5)
F : 600# (ASME B16.5)
G : 300# (ASME B16.5)
H : 150# (ASME B16.5)
J : 125# (ASME B16.1)
X : Special as Designated
(2) PIPING MATERIAL
A : ALLOY STEEL (Cr-Mo)
B : CARBON STEEL (CS)
C : STAINLESS STEEL (304 or 304L)
D : STAINLESS STEEL (316 or 316L), DUPLEX STAINLESS STEEL
E : CAST IRON or DUCTILE IRON (CI or DI)
F : GALVANIZED CARBON STEEL
G : COPPER
H : PVC or CPVC
J : PVDF
K : PLASTIC LINED CARBON STEEL
L : HIGH DENSITY POLYETHYLENE (HDPE)
M : POLYPROPYLENE (PP)
R : FRP, GRP, GRE, GRV, etc.
S : CONCRETE
(3) SERIAL NO.

TYPICAL IDENTIFICATION SYSTEM FOR
PIPE LINE, VALVE, EQUIPMENT AND FILTER, SCREEN,STRAINER, etc.

TAG NUMBERING SYSTEM IS BASED ON THE KKS NUMBERING SYSTEM. (DOC. NO.: 19K9-00GEN00-G02-0002)

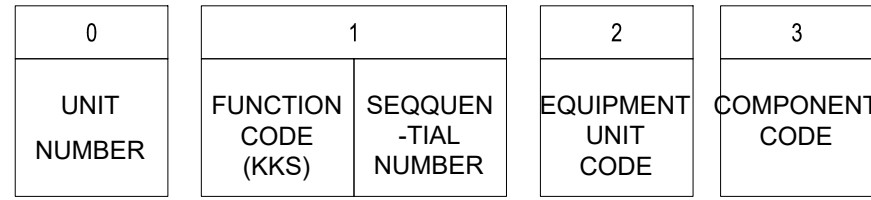


UNIT IDENTIFICATION NUMBERS

	PLANT UNIT	"UU" CHARACTER DESIGNATION
POWER PLANT	POWER PLANT COMMON	0
	FIRST GROUP COMMON (INCL. FIRST GROUP STG)	10
	FIRST GROUP #1 GTG / HRSG	11
	FIRST GROUP #2 GTG / HRSG	12
	SECOND GROUP COMMON (INCL. SECOND GROUP STG)	20
	SECOND GROUP #1 GTG / HRSG	21
	SECOND GROUP #2 GTG / HRSG	22

THE BASIS FOR THE DESIGNATION OF INSTALLATIONS AND SUBSYSTEMS
IN POWER PLANTS.

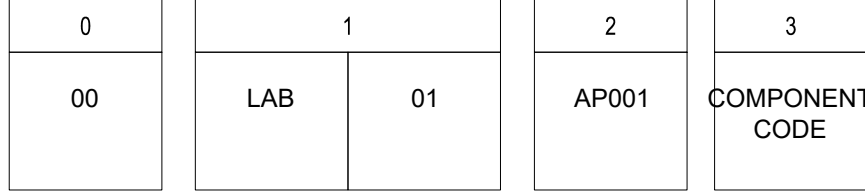
SERIAL NO. OF
BREAKDOWN LEVEL



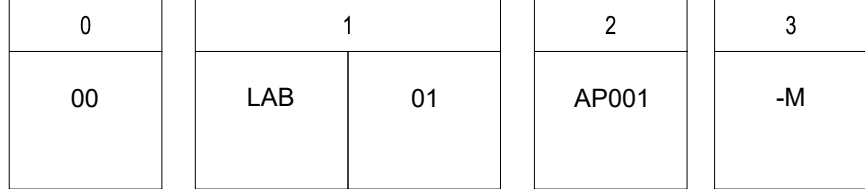
DESIGNATION OF
BREAKDOWN LEVEL

EXAMPLE OF
APPLICATION:

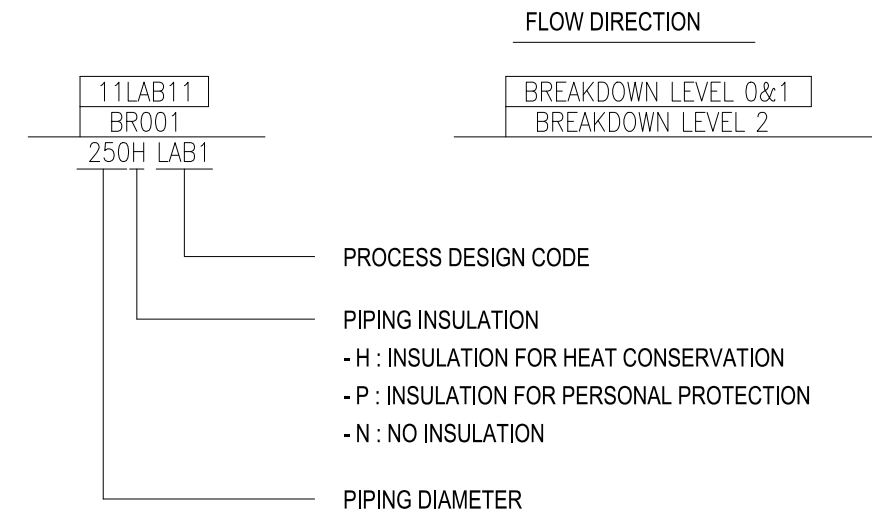
20LAB31AP001 (COGENERATION FEEDWATER PUMP A)



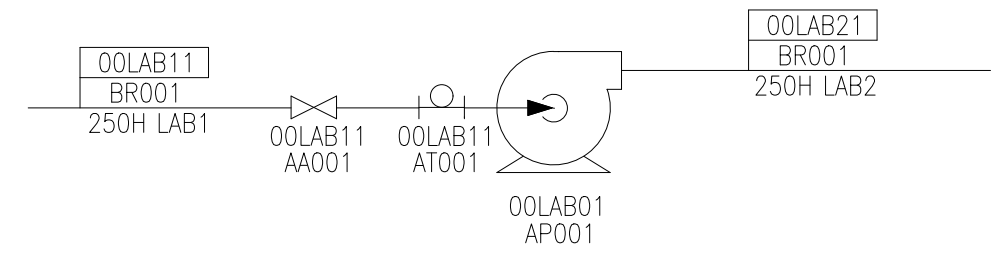
20LAB11AP001-M (COGENERATION FEEDWATER PUMP A MOTOR)



PIPING DESIGNATION FLAG
(POINTING IN FLOW DIRECTION INCLUDING PIPING DIAMETER,
PROCESS DESIGN CODE, PIPING INSULATION)



EXAMPLE FOR LINE VALVE, EQUIPMENT AND STRAINER IDENTIFICATION



GENERAL NOTES :

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NOTES :

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EMPLOYER:					

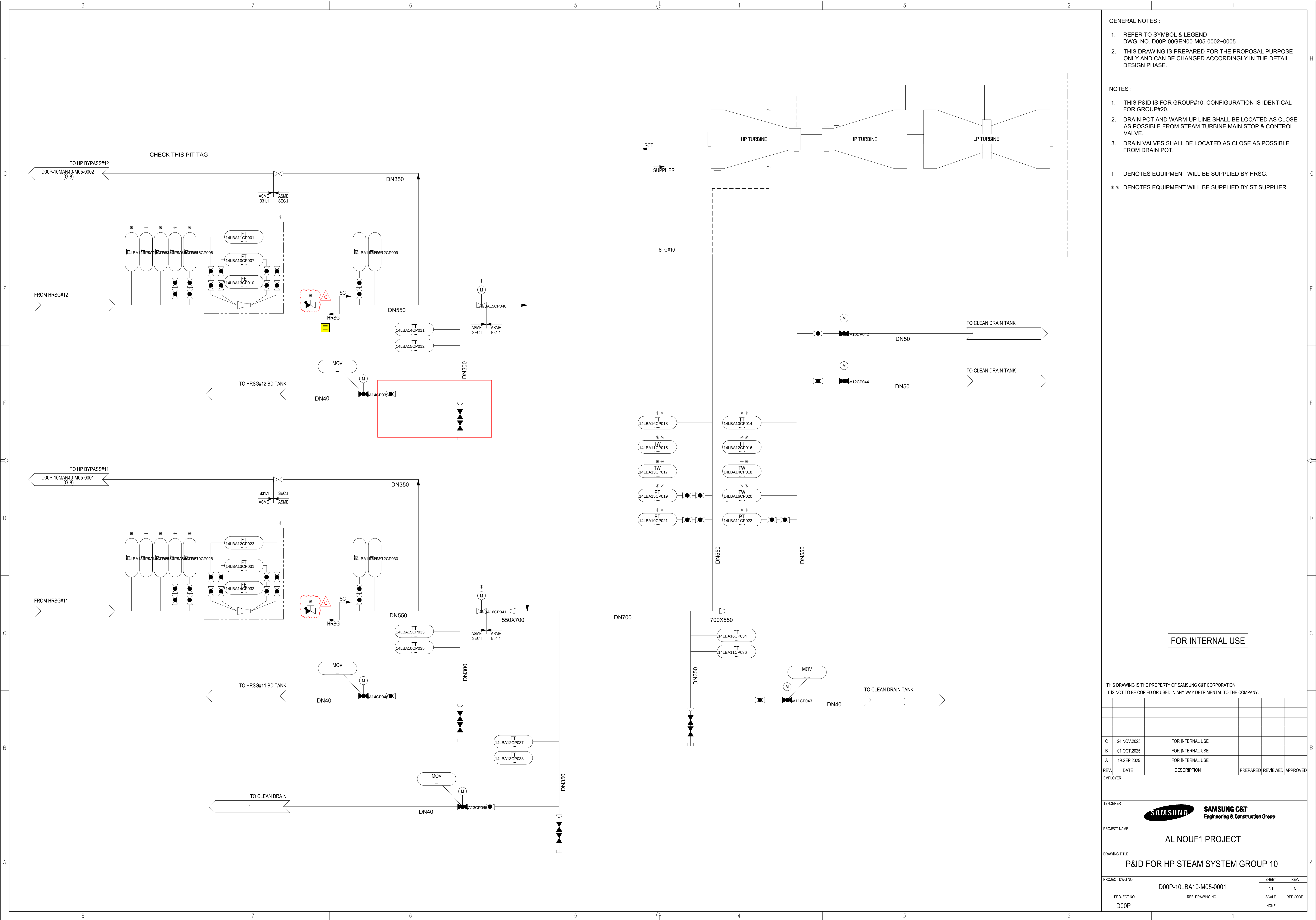
TENDERER
SAMSUNG **SAMSUNG C&T**
Engineering & Construction Group

PROJECT NAME
AL NOUF1 PROJECT

DRAWING TITLE
P&ID FOR SYMBOLS & LEGENDS 4

PROJECT DWG NO.	D00P-00GEN00-M05-0005	SHEET	REV.
PROJECT NO.	REF. DRAWING NO.	1/1	A
D00P		NONE	

NUMBER RANGE	VALVES	PIPE BR	INSTRUMENTATION CIRCUITS
001-100	-VALVES IN MAIN PIPES -VALVE GENERAL	-MAIN PIPES	-TRANSMITTER
101-190	-CONTROL VALVES / DAMPERS		-SWITCH
191-199	- SAFETY VALVES	-SUCTION AND PRESSURE RELIEF LINES ON SAFETY VALVES	
201-250	-TCE IS USING FOR MOV IN MAIN LINES	-NOT USED	-FOR TESTING EQUIPMENTS
251-299	-SHUT OFF VALVE MAIN LINE		
301-399	-INSTRUMENT ISOLATION	-PRESSURE LINES FOR INSTRUMENTS	-NOT USED
401-499	-VALVES IN DRAINS AND FLUSHING PIPES	-DRAINS, FLUSHING PIPES, PRESSURE SUPPERSSION PIPES	-ACCEPTANCE TESTS -TEST THERMOWELL
501-599	-VALVES FOR VENTING	-VENTING	-LOCAL MEASURING CIRCUITS -GAUGES
601-699	-VALVES FOR SAMPLING AND DOSING	-SAMPLING AND DOSING PIPES	-FLOW ELEMENT
701-799	-VALVES FOR INTERNAL CONTROLS	-FOR INTERNAL CONTROLS	-GROUPED
801-899	1. TCE IS USING FOR MOV IN DRAIN LINES 2. STEAM TRAP	-NOT USED	-GROUPED
901-999	-TCE IS USING FOR MOV IN VENT LINES	-NOT USED	-GROUP MEASURING CIRCUITS (E.G. MULTI POINT SELECTOR SWITCHES, MULTI CHANNEL RECORDERS)



- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND
DWG. NO. D00P-00GEN00-M05-0002-0005
 - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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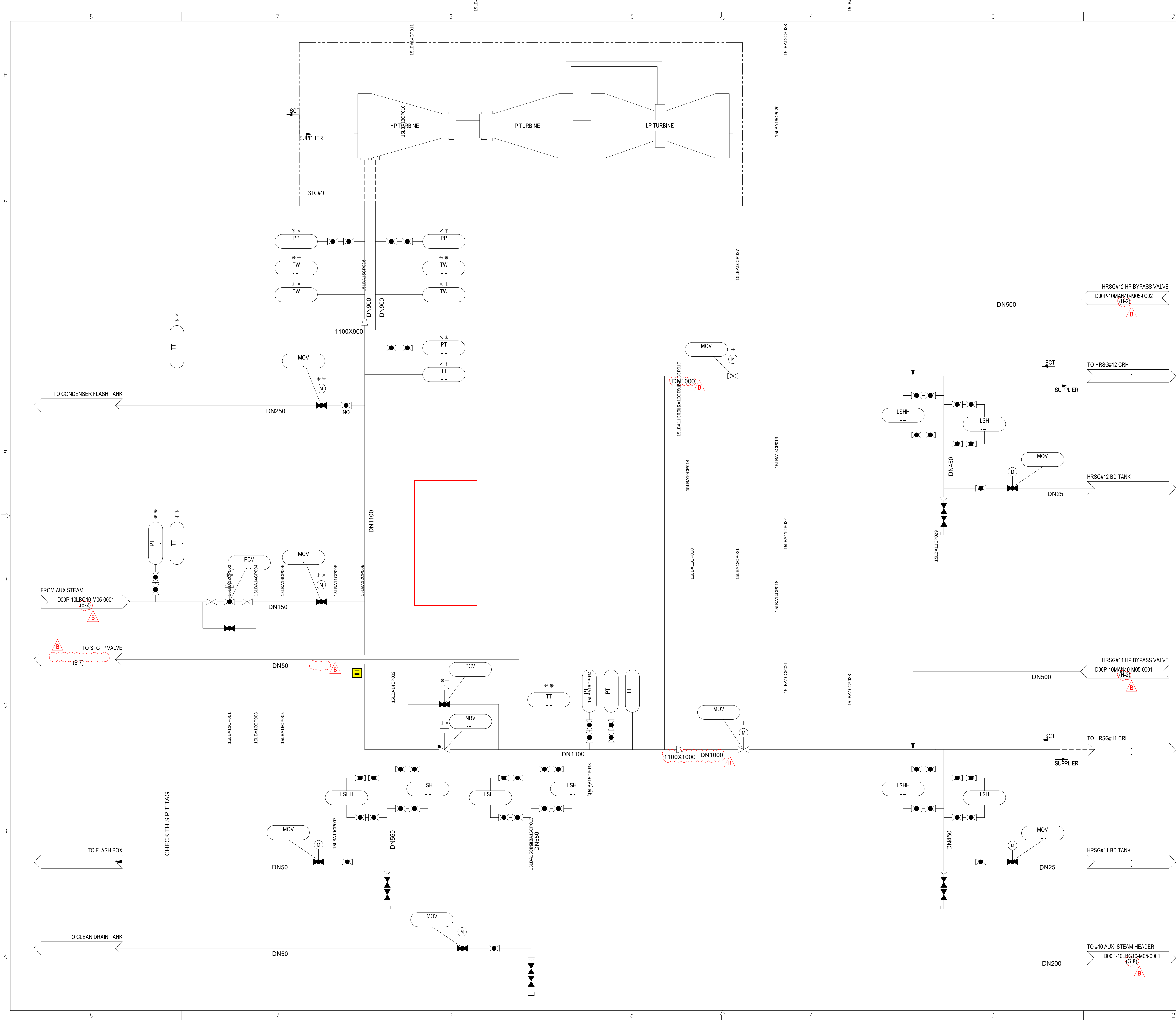
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
C	24.NOV.2025	FOR INTERNAL USE			
B	01.OCT.2025	FOR INTERNAL USE			
A	19.SEP.2025	FOR INTERNAL USE			

TENDERER
SAMSUNG **SAMSUNG C&T**
Engineering & Construction Group

PROJECT NAME
AL NOUF1 PROJECT

DRAWING TITLE
P&ID FOR HP STEAM SYSTEM GROUP 10

PROJECT DWG. NO.	PROJECT NO.	REF. DRAWING NO.	SHEET	REV.
D00P			1/1	C
			SCALE	REF. CODE
			NONE	



- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND
DWG. NO. D00P-00GEN00-M05-0002~3
 - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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A	19.SEP.2025	FOR INTERNAL USE		
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED
EMPLOYER			APPROVED	
TENDERER				
SAMSUNG SAMSUNG C&T Engineering & Construction Group				
PROJECT NAME				
AL NOUF1 PROJECT				
DRAWING TITLE				
P&ID FOR CRH STEAM SYSTEM GROUP 10				
PROJECT DWG NO.				SHEET
D00P-10LBC40-M05-0001				1/1
PROJECT NO.				SCALE
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REF. DRAWING NO.				REF. CODE
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